Split-track and duplicate treatment, ana581 default, page 1: how it was decided -- every ana578 conclusion re-tested, one mechanism at a time

step	the question, and the test that decided it (ana581)	decision vs ana578 (green = holds, orange = changed)
WHAT CHANGED in the tracks	ana581 = ana578 runs 79510-79516 with the DST repair AND new tracking (same TFs, Collect trees compared directly): +2.4% tracks per event, +0.9% after level-0 (pass 0.847 -> 0.834); fit-quality tail halved; charge balance +/- 1.024 -> 1.074; K^0_S mass -8.5 -> -2.9 MeV from PDG, V0 signal +25-35%. So nothing is carried over. sec 1	re-test every mechanism from scratch
CLEAN SAMPLE of events	Good TPC segments (bad-segment list over 7 runs), ZB crossings outside NearX0, one vertex per crossing, then the event cleaning: TFdup (a collision read again in an overlapping trigger frame, TF) and XTF (one track copied into another crossing of the same TF). Test: the like-sign MIXED spike at (dy, dϕ) = (0,0): x7.34 / x7.49 (nothing on) -> x1.18 / x1.11 (TFdup) -> 1.02 ± 0.004 (+ XTF, mixing exclusion). sec 2	TFdup: holds (13.0% of the rows; ana578 12.8%) XTF: holds, in-time INTT rule kept: in-time 0.946 right vs quality 0.923 on the same pairs (ana578 0.941 / 0.910)
LEVEL-0 CUTS events, tracks	Events: |vtx x|, |vtx y| < 0.5 cm, $\|z_{vtx}\|$ < 16 cm (8 cm in Finalize). Tracks: 0.1 < p_T < 20.1 GeV/c, ntpc $\geq$ 18, |dcaxy|, |dcaz| $\leq$ 1.5 cm. The per-crossing N-1 test needs AllX files: skipped (user, 11-02).	unchanged; not re-tested on ana581
AREA A single-track acceptance	A1 SiPhi zones: vertex-ϕ holes with large |dcaxy| (track density and <|dcaxy|> vs vertex ϕ). A2 central-membrane (CM) hole: tracks that cross z = 0 inside the TPC. Test: the bump of pion R_2(y,y) at |y| < 0.2 vs the mask width, decided after Area B (as for ana578). The hole is shallower again (minimum 0.52 of the plateau; ana578 0.42), but $\pi^-\pi^-/\pi^+\pi^-$ keep falling with width. A1, A2	A1: holds -- [70,83) only (2.26% of the tracks; ana578 2.3%); [94,115) still no hole A2: CHANGED -- w in [-0.09, 0.16) (8.0% of the pions; ana578 [-0.07, 0.10), 4.3%); user, 11-03
AREA B one particle, two tracks	B1 like-sign split band: |dϕ| ~ 0.7° x |1/p_{T1} - 1/p_{T2}|; NARROWER at small curvature difference (q98 0.46° vs 0.97°); 99.65-99.8% inside the pregate. B2 π^+ splits more than π^-: 1.77 ± 0.02 (ana578 1.80). B3 SL / SiSplitScore / RadialGap / MVTX match, one path off at a time (bump, survivors). B4 ULS veto: 7% more shared-silicon $\pi^+\pi^-$ pairs than ana578; C(Q)/(1+r_2) at 25-30 MeV 4.05 -> 1.15. B5 looper veto: ~15% fewer loopers. Ridge bump: like-sign 0.53 / 0.35 -> 0.024 / 0.019 (LS paths), unlike-sign 0.247 -> 0.016 (ULS veto); each cleaner only its own sign.	B1, B2, B4: hold (pregate, ULS veto SiSplitScore $\geq$ 0.05) B3: holds -- all 3 paths kept: MVTX path NEEDED, RadialGap nearly redundant (89 extra tracks) B5: holds at 0.08; ~22% of the looper excess lies above the cut (watch)
AREA C two real tracks, lost or merged	C1 tsep (LS two-track merging): LS C(Q) with off / 2.0° / 3.0°; S/M vs min |dϕ^*| (0.85 at 0-0.5°). C2 isep (near-vertex loss, LS and ULS): S/M vs |dϕ^*(3 cm)| (LS 0.60, ULS 0.75 at 0-0.5°, back to 1 by 2°); the missing same-event pairs inside a cut of width w peak at exactly w = 2.0°. C1, C2	C1: holds (|dy| < 0.06, min |dϕ^*| < 2.0°); user C2: holds (|dy| < 1.0, |dϕ^*(3 cm)| < 2.0°); user
CROSSING CORRECTION and AREA D	Crossing correction (not a cut): pairs added, ana581 vs ana578ABC: $\pi^+\pi^-$ 1.90 vs 1.96 e-3; ~85% is the isep acceptance fill, the real ULS loss 1.10 ± 0.37 vs 1.27 ± 0.20 e-3 pairs/event; flat in event rate. D1 the ±15° ULS notch: S/M vs meeting radius 0.94, 0.93, 0.93, 0.95 (0.10-0.30 m): the same gap loss.	correction: the same as ana578 D1: the same -- monitored, not cut
V0 RULES	In-peak V0s claim their daughters; $\Lambda/\bar{\Lambda}$ daughters are split partners. Re-test: ana581 vs ana581_nodau: check off = big low-Q C(Q) and dϕ ~ 0 R_2 excess in every track-Λ type; track-K^0_S identical. page 5	holds -- the partner check stays on

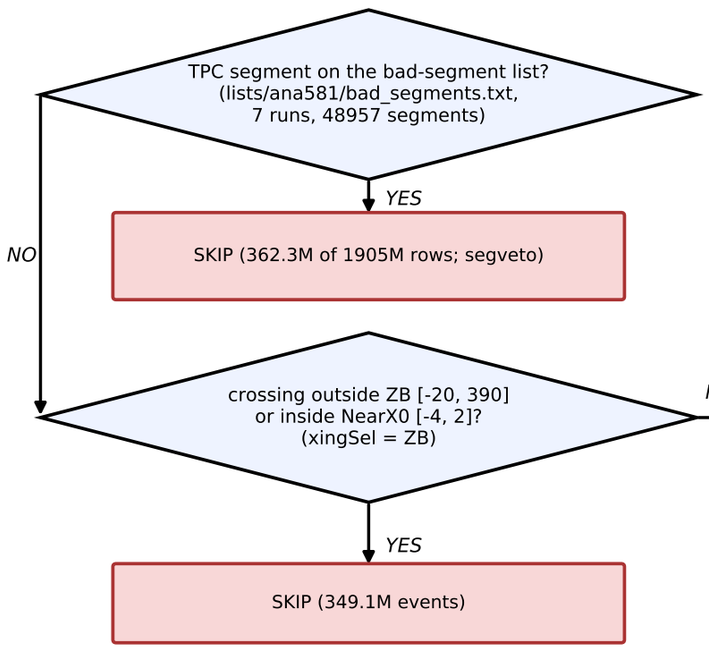
RESULT: the ana578 procedure with the ana578 cut values, except one: the CM mask is wider, [-0.09, 0.16). The new tracking changed the SIZE of several mechanisms -- 7% more unlike-sign shared-silicon pairs, a narrower like-sign split band, ~15% fewer loopers, a shallower CM hole -- but not the mechanisms. Each was re-tested with the symptom it was made for, on the clean output of the step before, so each cut was judged alone; the same cuts came back, and the CM width was re-chosen because the like-sign bump vanishes at [-0.05, 0.08) while the unlike-sign and $\pi^-\pi^-$ bumps keep falling.

Numbers: 7-run production ana581 (189 chunks: Finalize run record, chunk logs) for fractions of rows or events; else the run-79516 tests (10M events or all of it, 152M after cleaning) in README_SplitTracks581 (section at the end of each box). Pages 2-5: the flowcharts.

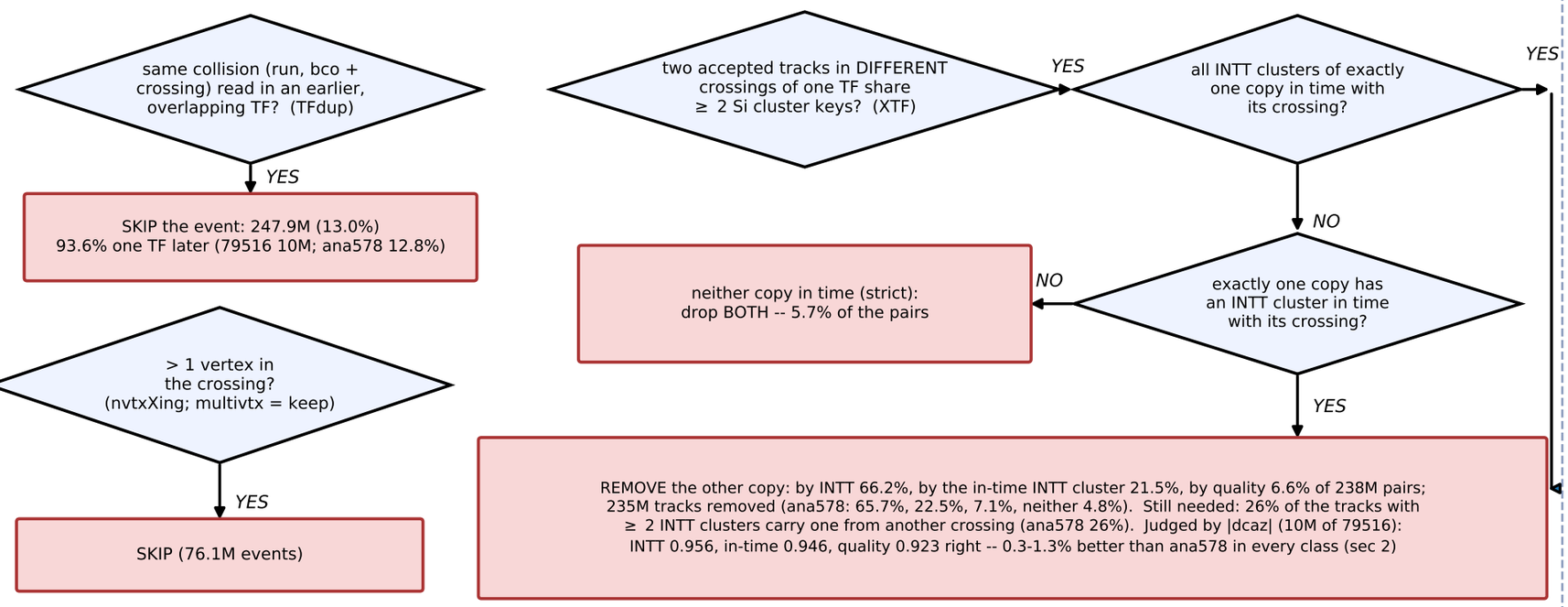
Split-track and duplicate treatment, ana581 default, page 2: the clean sample of events, the level-0 cuts, Area A (single-track acceptance)

CLEAN SAMPLE -- which events can be used at all (event = one crossing with a vertex; the DSTs hold many events per trigger frame, TF)

(a) the event selection (ZB)



(b) event cleaning: pre-passes before the event loop

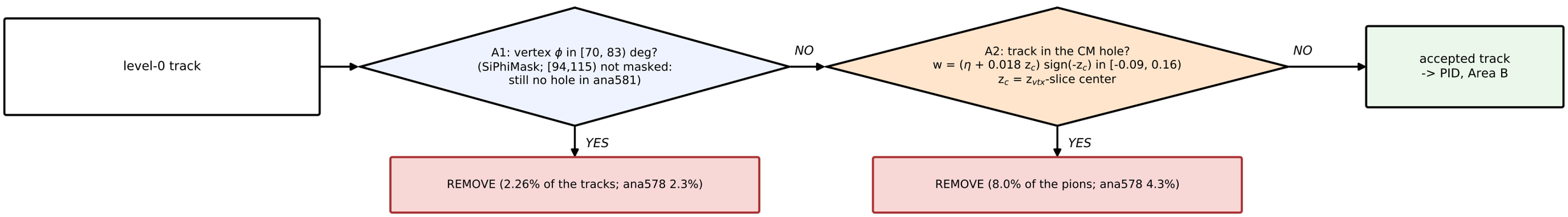


LEVEL-0 CUTS -- per event and per track (AcceptEvent, AcceptTrackBase)

AcceptEvent: |vtx x|, |vtx y| $\leq$ 0.5 cm, $|z_{vtx}| < 16$ cm in 16 slices (physics: $|z_{vtx}| < 8$ cm, Finalize)
AcceptTrackBase: $0.1 < p_T < 20.1$ GeV/c, ntpc $\geq$ 18, |dcaxy|, |dcaz| $\leq$ 1.5 cm
859M events reach the event loop, 532M at $|z_{vtx}| < 8$ cm

Not re-tested on ana581 (the per-crossing N-1 pages need AllX files;
user skipped them, 11-02: the crossing cut did not change).
83.4% of the Collect tracks pass (ana578 84.7%): the extra ana581 tracks
mostly fail level-0 (net +0.9% per event; sec 1). Cuts unchanged.

AREA A -- single-track acceptance (AcceptTrack): masks for holes that change from event to event

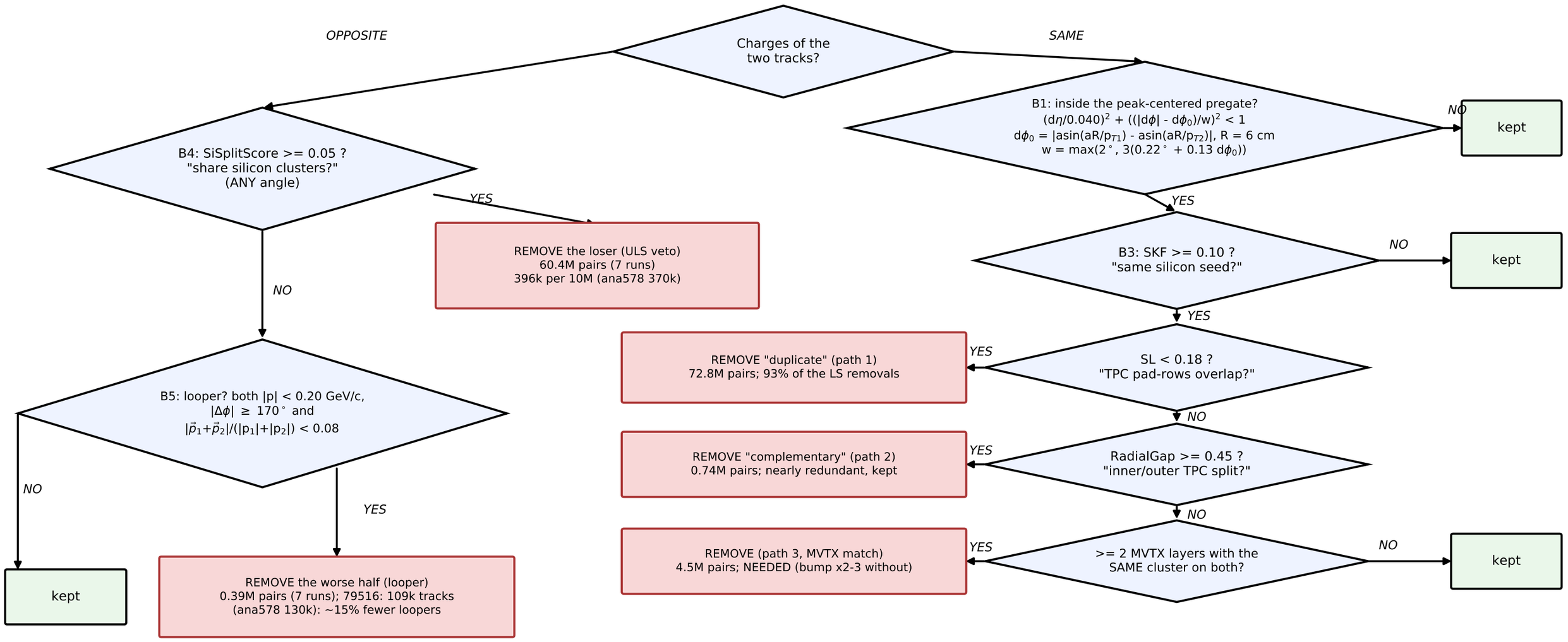


A1 (SiPhi_ana581, 2.85e9 / 2.67e9 tracks per charge): [70,83) still a hole with large |dcaxy| (2.2-2.3% of the tracks vs 3.6% flat, <|dcaxy|> 1.6-1.7x the median), as ana578.
A2 (ST581_A2_CMMask, width scan on all of 79516 with every B cleaner): mask off leaves a 0.015-0.020 bump; $\pi^+\pi^+$ is gone from [-0.05, 0.08), while $\pi^-\pi^-$ and $\pi^+\pi^-$ keep falling slowly to the widest window. User chose [-0.09, 0.16): bumps 0.0031 / -0.0004 / 0.0012 ($\pi^+\pi^-$ / $\pi^+\pi^+$ / $\pi^-\pi^-$).
Why after B: the y ~ 0 bump with no track cleaner is mostly split tracks (like-sign 0.53 -> 0.02 with the LS paths), so the CM hole is judged after them (page 3).

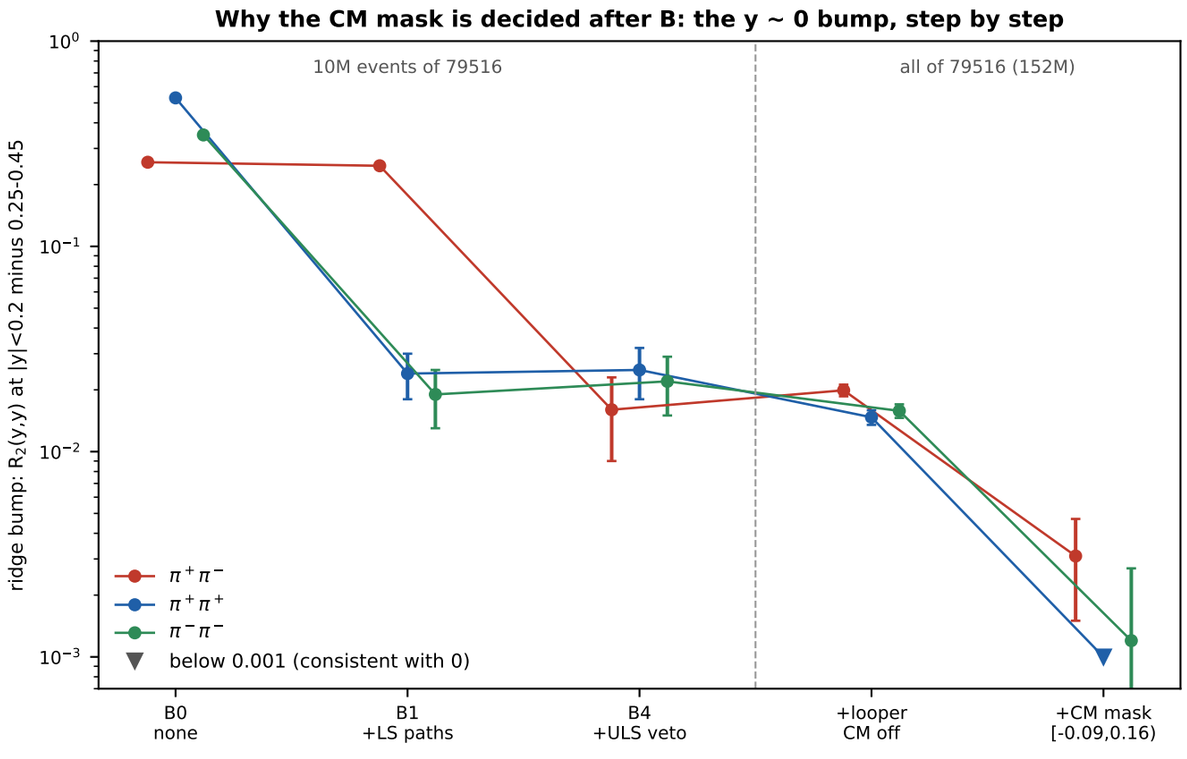
Red = removed / skipped in the ana581 default. Event numbers: 7-run production ana581 (Finalize run record; skips counted in order: TF copies, segments, crossing, multi-vertex). Track fractions: README_SplitTracks581 secs 1, 2, A1, A2.

Split-track and duplicate treatment, ana581 default, page 3: Area B -- one particle, two tracks (per event, every pair of accepted charged tracks)

AREA B -- the split-track pre-pass (corral_loop.cxx), before any pair is formed; tested in steps B0 (none) -> B1 (+LS) -> B4 (+ULS veto) -> B5 (+looper)



loser rule: fewer TPC clusters, then worse fit quality (looper: then the larger |dca| sum). 79516, 10M: 1.021M tracks flagged by the LS paths (ana578 1.073M).
All of Area B, 7 runs: 227M tracks flagged, 20.8% of the events have at least one flag (ana578 20.2%). A removed track leaves singles, sibling and mixed pairs alike.



Each cleaner removes the bump of its own sign only: the LS paths take $\pi^+\pi^+$ 0.53 -> 0.024 and $\pi^-\pi^-$ 0.35 -> 0.019, the ULS veto takes $\pi^+\pi^-$ 0.247 -> 0.016.
Left after B (all of 79516): a COMMON bump of 0.015-0.020 in all three -- not a split-track signature. The CM mask takes it to -0.0004..0.0031: that residual is the CM hole (A2, page 2).

The same B tests, ana581 vs ana578:
B1 same 0.7° x $|1/p_{T1} - 1/p_{T2}|$ law; band narrower at small curvature difference (q98 0.46° vs 0.97°); 99.65-99.8% of the shared-silicon pairs in the pregate
B2 π^+/π^- splits 1.77 ± 0.02 (ana578 1.80)
B3 without path 3: +30k tracks, bump +0.047 / +0.024; 760 in-gate survivors (ana578 764)
B4 7% more shared-silicon $\pi^+\pi^-$ pairs: the larger excess at 0 of step 2
B5 63k looper pairs over a control of 28.8k below 0.08 (ana578 75k); ~22% above (watch)

Pair counts: 7-run production ana581 (chunk logs). Else run 79516 (B0-B4 10M of list_00, B5 and A2 all of 79516; README_SplitTracks581 secs B1-B5, A2). B0-B4 bumps with the CM mask off.

Split-track and duplicate treatment, ana581 default, page 4: Area C (two real tracks), the crossing correction, Area D and the watch list

AREA C -- pair level (CalcRm::PairInfo): two real tracks, lost or merged; sibling AND mixed pairs alike, in this order; then the maps

C1: LS charged pair with $|dy| < 0.06$ and min over R = 0.30-0.78 m of $|d\phi^*| < 2.0^\circ$? (two-track resolution, tsep)

NO

C2: charged pair with $|d\phi^*(R = 3\ cm)| < 2.0^\circ$ at $|dy| < 1.0$, LS or ULS ? (near-vertex two-track cut, isep)

NO

Mixed EVENT pair from adjacent trigger frames? (mixing exclusion, with XTF)

YES

REJECT the pair
LS pions: 0.36-0.37% of sibling / 0.36-0.37% of mixed pairs (ana578 0.36-0.38% / 0.36%)

YES

REJECT the pair
ULS pions 0.80% S / 0.87% M; LS 0.82-0.83% / 0.86% (ana578 0.82 / 0.89; 0.84 / 0.88)

YES

SKIP the event pair
same-TF share of the mixed pion pairs left: 4.0% (ana578 5.1%)

then the (dy, dϕ) maps get the CROSSING CORRECTION (pt ordering; the dirty dϕ side <- its clean mirror). A correction, not a removal:
+1.90e-3 of the $\pi^+\pi^-$, +1.4-1.5e-3 of the like-sign pairs (ana578ABC 1.96, 1.3-1.4); ~85% of it fills the isep hole; flat in event rate (CrossCompare_ana581.pdf).

AREA D and the WATCH LIST -- monitored in every production, not removed

D1: the ±15° ULS notch (ST581_D1_Notch.pdf)
ULS depth ~0.98 (S/M in 10-20° / a local pol2 baseline at 4-8, 24-40°; not 1:1 with the ana578 ratio to 25-50°); LS 0.98-0.99.
vs the radius where the two tracks meet in ϕ, R_m:
0.94 (0.10-0.15 m), 0.928, 0.93, 0.95 (0.25-0.30), 0.98, then 1.02-1.31
= the same INTT-TPC-gap matching loss as ana573 and ana578.
Pairs that MEET in the gap: 0.936 (ana578 0.937). It does not shrink with dy (0.953 -> 0.913): the matching seems to use ϕ only -- a note for tracking. Deepest at $|dp_T|$ 0.2-0.3 GeV/c (0.92).
No cheap cut: monitored.

Watch items
- the dϕ = 0 dip in LS R_2 ($|dy| < 0.02$) left after tsep: 0.28 vs 0.36 for the neighbors, with 2.0° or 3.0° (C1, open)
- the looper excess above the 0.08 cut, ~22% (B5)
- crossing correction: $\pi^+X^- < \pi^-X^+$ as in ana578 (not understood); like-sign fractions 5-10% above ana578ABC
- charge balance +/- 1.074 at level-0 (ana578 1.024); the extra tracks that fail level-0 (sec 1)
- V0: K^0_S window 488.4 ± 22.8 MeV, peak now ~494.7 MeV (inside); $\pi^-\Lambda$, $\pi^+\bar{\Lambda}$ C(Q) ~1.2 at 20-40 MeV with the check on (2σ)
- tracks removed per cleaner: the chunk logs and Finalize

Red = removed / skipped in the ana581 default. Pair-cut fractions and same-TF share: 7-run production ana581 (counters of the 189 chunk logs, pions). Notch and C1/C2: all of 79516 (README_SplitTracks581 secs C1, C2, D1; crossing step).

Split-track and duplicate treatment, ana581 default, page 5: V0s -- which V0s pair, and what happens to their daughters

STAGE V1 -- per event, every KFP V0 candidate (K^0_S, Λ, $\bar{\Lambda}$)

V0 candidate
(zero-sentinel eta/phi/mass/ctau
rebuilt, README_v0etaSpike)

within $\pm 3\sigma$ of its mass peak?
K^0_S 0.4884 ± 3(0.0076), Λ 1.1143 ± 3(0.0035)
$\bar{\Lambda}$ 1.1134 ± 3(0.0028) GeV
(V0PEAK_*, fluct_common.h; ana573 fits)

YES → IN PEAK: enters the V0 pair types and
CLAIMS its two daughters
K^0_S 94.4%, Λ 53.7%, $\bar{\Lambda}$ 43.4% of the candidates
(ana578 84.6, 40.8, 29.3%: less background now)

NO → OFF PEAK: not paired (it would pair with its own
daughter); its daughters stay ORDINARY tracks

STAGE V2 -- per event, the in-peak daughters in the track machinery (corral_loop.cxx)

in-peak daughter: OUT of every track pair type,
OUT of the split-track candidate list,
so NEVER a split-track loser

parent V0?

Λ, $\bar{\Lambda}$ → Λ, $\bar{\Lambda}$: another accepted track
is a SPLIT COPY of a daughter?
same charge: LS paths and $\Delta R < 0.05$
opposite charge: SiSplitScore $\geq$ 0.05;
a proton-tagged partner of the (anti)proton
daughter is its wrong-sign copy

YES → REMOVE the other track
(the daughter wins)
7 runs: LS 159,286,
ULS 130,031 partner pairs

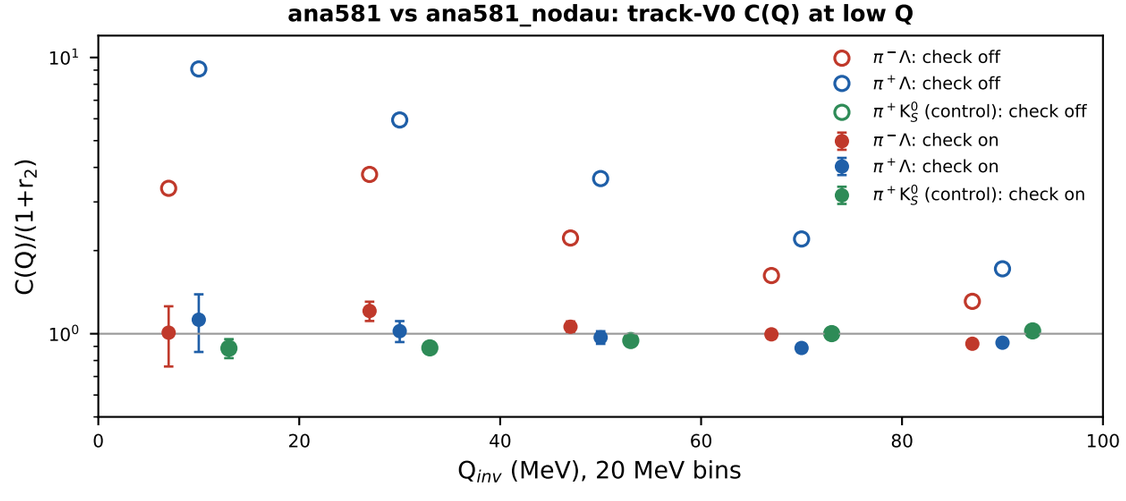
K^0_S →

K^0_S: NO partner check
(ana573: it cut real $\pi^\pm K^0_S$ pairs)

Rules from ana573 (README_CQcomparison sec 2.2): the partner check removed the pΛ and $\pi\Lambda$ excesses at Q < 0.2 GeV; checking K^0_S daughters too cut ~10% of the real $\pi^\pm K^0_S$ sibling pairs at Q < 0.4 GeV.

RE-TEST ON ana581 (2026-11-05) -- ana581 (default) vs ana581_nodau (partner check off), same 189 chunks, 532M events

Off: OFF/ON singles +0.004-0.014% (tracks only; V0s identical).
$1+r_2$ of every track-$\Lambda/\bar{\Lambda}$ type +0.8-3.6%; all other
types within 1e-4.
$C(Q)/(1+r_2)$ at Q < 0.1 GeV: big excesses with the check off
in EVERY track-Λ type, all charge combinations (the
unlike-sign rule too): $\pi\Lambda$ 1.0-1.3 -> 3-9, pΛ 20-40 MeV
3.2 ± 2.2 -> 14.8. $R_2(d\phi)$ at small dy: up to 2-3x in
the $d\phi \sim 0$ bins; on, they match the 30-60° neighbors.
Control: track-K^0_S identical in $1+r_2$, C(Q), R_2.
Decision (user OK, 11-05): the check stays on. V0DaughterCompare_ana581.pdf



Green = kept / paired / holds, red = removed, yellow = watched. V0 numbers: 7-run production ana581 (chunk logs). Mass windows unchanged from ana573.
Data panel: Finalize ana581 and ana581_nodau; errors drawn for check on only (same events: check off differs by far more than its errors).